Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026, 7:59 AM (MYT)

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

Kopi-Paste Coders

Project / Solution Name

ESG-Passport

Name (Leader)

Lee Boon Yew

Email & Phone (Leader)

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University Affiliation(s)

Tunku Abdul Rahman University of Management and Technology (TARUMT)

Live Prototype URL

https://esg-passport.replit.app/

Code Repository URL (public / access granted to judges)

https://github.com/nestchao/ESG-Passport

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The Next.js browser interface calls a same-origin server proxy, which adds a server-only backend key before forwarding requests to FastAPI. PostgreSQL stores records, revisions, questions, factors, approvals and audit events; private Supabase Storage retains source documents. POST /documents invokes the backend OCR adapter. Offline mode uses RapidOCR/PDF text extraction. Explicit remote modes call Gemini or Grafilab for transcription, then a deterministic parser supplies unverified JSON candidates. Confirmed activities alone feed the profile. Question mapping imports the shared ai.workflow mapping rules. Evidence adapters import reviewed-form, matching and coverage helpers. Existing calculation rules produce displayed monthly metrics; shared preparation/approval helpers additionally retain gaps at owner sharing. The two calculation paths are not yet fully unified. Owner approval is bound to the draft and selected evidence versions, with explicit gap acknowledgements. The immutable share snapshot exposes selected attachments through token routes. Replit Core (organiser-provided) hosts the frontend and backend. Browser OCR is a separate on-device path and does not save a backend record. Live Orbyte access is not implemented. The staged persistent OCR router is not used by the hosted upload adapter.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Next.js 16 / React 19 / TypeScript >=5.7	Frontend; versions recorded in package-lock.json	MIT / MIT / Apache-2.0
Python 3.12 / FastAPI >=0.115 / SQLAlchemy >=2.0.38 / Psycopg >=3.2 / Alembic >=1.15 / Uvicorn >=0.34	Backend API, SQL data access, PostgreSQL driver, schema migrations and ASGI server	Python: PSF; FastAPI: MIT; SQLAlchemy: MIT; Psycopg: LGPL-3.0; Alembic: MIT; Uvicorn: BSD-3-Clause
Pydantic / HTTPX >=0.28 / python-dotenv >=1.0	Validation, API transport, local settings	MIT / BSD-3-Clause / BSD-3-Clause
RapidOCR >=1.4 / ONNX Runtime / NumPy	Offline OCR; backend and private versions differ	Apache-2.0 / MIT / BSD-3-Clause
Pillow >=11 / pypdf >=5 / pypdfium2 >=5	Image and PDF handling	Pillow: MIT-CMU; pypdf: BSD-3-Clause; pypdfium2: Apache-2.0 / BSD-3-Clause; PDFium: BSD-style + bundled third-party notices
Tesseract.js >=7 / PDF.js >=5.7 / jsonschema 4.26.0; Tailwind CSS >=4 / lucide-react >=0.468	Browser OCR/PDF and workflow validation; Tailwind, lucide, uv and Codex PDF tools	Tesseract.js / PDF.js: Apache-2.0; jsonschema / Tailwind: MIT; lucide-react: ISC; other tools: see lockfiles

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API	Recorded OCR comparisons; optional upload OCR	Free tier
Grafilab inference API	Qwen vision comparison and optional fallback	Paid credit; user-reported RM80
Supabase Storage REST API	Private originals and token-mediated downloads	Free
Supabase Postgres connection	Persistent application data (SQL, not REST)	Free
OCR asset distribution services	Tesseract worker/language assets; browser only	Public downloads; upstream terms

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Core	Frontend and backend hosting; organiser-provided deployment	Organiser-provided Replit Core; Asia
Supabase	Postgres and private Storage	Free; Northeast Asia (Tokyo), ap-northeast-1
Google AI Studio / Gemini	Hosted inference; no model training	Free-tier user preference; provider terms apply
Grafilab	Hosted Qwen inference; saved usage evidence	RM80 credit user-reported
GitHub	Code hosting, history and CI configuration	Free plan; no paid features

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gemini-3.5-flash-lite	Google Gemini	Recorded pre-trained OCR comparison	Gemini API terms
gemini-3.1-flash-lite	Google Gemini	Configured pre-trained backend OCR model	Gemini API terms
qwen3.5-397b-a17b	Grafilab	Recorded pre-trained vision comparison	Grafilab API terms; weight terms distinct
PaddleOCR ONNX models	RapidOCR distribution	Pre-trained CPU OCR; exact files in manifest	PaddleOCR/RapidOCR model weights: Apache-2.0; ONNX Runtime: MIT
English traineddata	Tesseract.js / tessdata	Pre-trained browser text recognition	Upstream Apache-2.0 notices

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
synthetic data 360 images (demo/eval)	github.com/ nestchao/ ESG-Passport	Synthetic; team-created	Apache-2.0	N

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Codex desktop; GPT 6.1 Sol	Code, tests, synthetic generator, research and documentation	Major AI assistance; team review required
Claude Opus 5.5	Desk research and fact-checking support; drafting the PRD, pitch deck and video script; scripted screen recording of the live demo (Playwright) and slide animations; generating a fictional sample bill for the demo	Major for pitch materials and demo video; all content reviewed and edited by the team
Luvvoice (AI text-to-speech)	Voice-over narration for the pitch video, generated from the team's script	Moderate; narration audio only, script written and checked by the team

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-assisted changes were checked against source documents, code and explicit rules. The recorded comparison used the same eight synthetic development images for both models, with frozen labels, field-level metrics, latency and saved responses. Twelve offline mapping/evidence bridge tests passed. OCR candidates still require manual confirmation; sharing requires Owner review and gap acknowledgements. Known defects and untested deployment scope are recorded. No perfect-accuracy or manual time-saving claim is inferred. The team must complete final review of licences, sponsor usage, originality, security and submission declarations before signing.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit (organiser-provided)	Replit Core subscription / platform	Frontend and backend hosting for the hackathon prototype

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

The organiser-provided Replit Core resource is used to host the frontend and backend of the ESG-Passport hackathon prototype. Hosting is necessary to make the application accessible to judges, but Replit itself is not essential to the system's core logic. The frontend, backend, Supabase database and external OCR integrations could be deployed on another compatible hosting platform with suitable configuration.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The demo is intended for team-created synthetic documents and fictional identifiers. The shared hosted workspace stores uploaded files, extracted text, quantities, questions, revisions, simulated reviewer names and audit events in Supabase. When remote OCR is enabled, document images are sent to Google or Grafilab; their terms and retention apply. Browser OCR processes selected documents locally, although OCR assets may be fetched externally. Credentials stay server-side and source storage is private. The public passwordless workspace uses simulated roles; visitors can view, edit and permanently delete eligible fictional records. Real or confidential information must not be uploaded. Records persist until explicitly removed; no automatic retention schedule is implemented. Approved evidence has deletion restrictions. A real Malaysian deployment requires PDPA review covering notice, permitted processing, access, security, retention and cross-border transfers, as well as actual authentication and tenant isolation. No production privacy certification is claimed.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The eight-image synthetic comparison is a development screen, not real-world validation. Models may omit units, add unsupported fields or misread degraded text. Both models returned one extra field; outputs require source review. API rate limits and outages remain possible. Known defects include a 60-second proxy deadline shorter than sequential remote fallback, estimated meter readings defaulting to non-estimated, and browser fuel receipts receiving a monthly period. These require re-testing before being claimed fixed. Mapping is rule-based with limited environmental coverage; compound or unsupported requirements need manual handling. Source and organisation boundaries are not independently verified. Existing profile calculations and stricter preparation checks remain separate. Persistent OCR routing and budget controls were recently added; hosted-path integration requires verification. The public workspace uses simulated roles rather than production authentication. Manual fallback, explicit gap acknowledgements and immutable snapshots reduce risk but do not establish assurance.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date]

25 October 2026